

Experiment 4: Missionaries and Cannibals Problem

Aim

To solve the Missionaries and Cannibals problem using Artificial Intelligence search techniques.

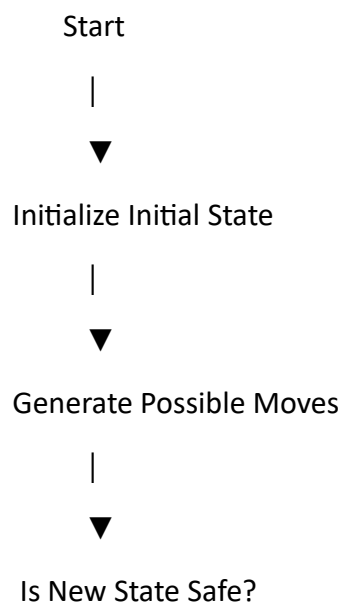
Objective

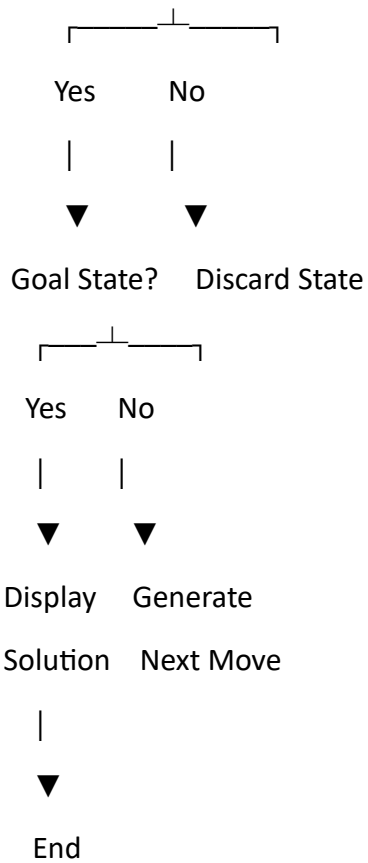
To safely transfer three missionaries and three cannibals from the left riverbank to the right riverbank without ever leaving missionaries outnumbered by cannibals on either bank.

Algorithm

1. Represent the initial state as (3 Missionaries, 3 Cannibals, Left Bank).
2. Generate all possible valid boat moves.
3. Check whether each new state is safe.
4. Ignore invalid or previously visited states.
5. Continue searching until the goal state (0 Missionaries, 0 Cannibals, Right Bank) is reached.
6. Display the sequence of valid moves.

Flowchart





Python Code

```
from collections import deque
```

```
def is_valid(m_left, c_left):
```

```
    m_right = 3 - m_left
```

```
    c_right = 3 - c_left
```

```
    if m_left < 0 or c_left < 0 or m_left > 3 or c_left > 3:
```

```
        return False
```

```
    if (m_left > 0 and m_left < c_left):
```

```
        return False
```

```
if (m_right > 0 and m_right < c_right):
```

```
    return False
```

```
return True
```

```
def missionaries_cannibals():
```

```
    start = (3, 3, 1)
```

```
    goal = (0, 0, 0)
```

```
    queue = deque([(start, [])])
```

```
    visited = set()
```

```
    moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
```

```
    while queue:
```

```
        state, path = queue.popleft()
```

```
        if state == goal:
```

```
            return path + [state]
```

```
        if state in visited:
```

```
            continue
```

```
        visited.add(state)
```

```
        m, c, boat = state
```

```
        for dm, dc in moves:
```

```
            if boat == 1:
```

```
        new = (m-dm, c-dc, 0)
    else:
        new = (m+dm, c+dc, 1)

    if is_valid(new[0], new[1]):
        queue.append((new, path+[state]))

solution = missionaries_cannibals()

for step in solution:
    print(step)
```

Sample Output

```
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
```

Result

Thus, the Missionaries and Cannibals problem was solved successfully using state-space search.